

michal-et-al-2021-when-and-why-do-people-act-on-flawed

Auto-assembled report. Section content is drawn from the summarization pipeline outputs.

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Paper Synthesis

Not available for this paper.

Abstract

Preserved from source without summarization.

Today's citizens are expected to use evidence, frequently presented in the media, to inform decisions about health, behavior, and public policy. However, science misinformation is ubiquitous in the media, making it difficult to apply research appropriately. Across two experiments, we addressed how anecdotes and prior beliefs impact readers' ability to both identify flawed science and make appropriate decisions based on flawed science in media articles. Each article described the results of flawed research on one of four educational interventions to improve learning (Experiment 1 included articles about having a tidy classroom and exercising while learning; Experiment 2 included articles about using virtual/augmented reality and napping at school). Experiment 1 tested the impact of a single anecdote and found no significant effect on either participants' evidence evaluations or decisions to implement the learning interventions. However, participants were more likely to adopt the more plausible intervention (tidy classroom) despite identifying that it was unsupported by the evidence, suggesting effects of prior beliefs. In Experiment 2, we tested whether this intervention effect was driven by differences in beliefs about intervention plausibility and included two additional interventions (virtual reality high plausible, napping low plausible). We again found that participants were more likely to implement high plausible than low plausible interventions, and that evidence quality was underweighed as a factor in these decisions. Together, these studies

suggest that evidence-based decisions are more strongly determined by prior beliefs than beliefs about the quality of evidence itself.

Keywords: Anecdotes, Evidence evaluation, Scientific reasoning, Education, Prior beliefs

Introduction (Paper-Level)

Not available for this paper.

General Discussion

Not available for this paper.

Critique

Not available for this paper.